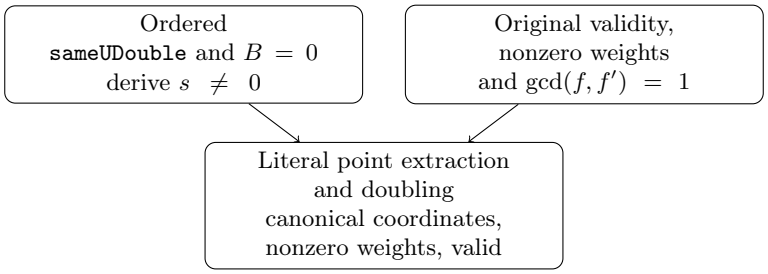
Ordered B-zero route: composed preservation

Producer result; independent fidelity review pending.

The exact ordered `sameUDouble` dispatcher and literal $B = 0$ derive $s \neq 0$. Composing that guard with the retained word-level point extraction and doubling theorem yields canonical extracted coordinates, a nonzero extracted ordinate, a selected doubled payload with canonical coordinates and nonzero weights, and normalized Mumford validity. Unlike the earlier payload theorem, this composition does not assume $s \neq 0$ separately.



Assumptions and statement boundary

Prime $5 \leq p < 2^{31}$; twelve canonical input coordinates; four canonical curve coefficients; degree-one arithmetic without lookup tables. The curve is the monic quintic without an X^4 term. Input a has normalized Mumford validity and nonzero z, w; the polynomial and its derivative are coprime. Input b supplies the exact crossed-coordinate dispatch; no additional validity or nonzero-weight premise about b is consumed.

This is the selected B-zero helper route. It does not formalize all other `add22` returns or prove a divisor-class-sum statement, compiler behavior, or operation counters. The imported BZero and word-guard producer packets retain their pending independent-review status.

Fixed original-input check

For $p = 5$, coefficients $(a_0, a_1, a_2, a_3) = (0, 4, 1, 0)$, choose

$$a = (1, 0, 4, 0, 2, 3), \qquad b = (4, 0, 1, 0, 1, 1)$$

in (q, r, s, t, z, w) order. They use different coordinate scales. The outer classifier returns `sameUDouble`, $B = 0$, the extracted (x, y, z, w) is $(4, 4, 3, 3)$ and the doubled output is

$$(q, r, s, t, z, w) = (2, 1, 3, 2, 3, 3).$$

The complete composed proposition is applied without an $s \neq 0$ argument. Original normalized validity and the explicit polynomial Bezout certificate are reused from the byte-preserved field fixture module.

Six new fixed propositions also check the earlier inverse branch for zero ordinates, a same-double input with $B = 3$ outside this lemma, and exclusion of $p = 2$. These are fixed proof controls, not a cohort or a timing experiment.

Evidence and next step

The main declarations, copied fixture module, new controls and separate axiom readback have individual command and input-binding receipts. `status.json` records their exact outcomes. `control-data.json` contains typed inputs and outputs; `flow.svg` is the editable diagram. No new cost ratio is assigned. Independent source and statement review precedes canonical integration.

Sources: retained `projective_law_gate.rs`, lines 727–753 and 911–935; `PROOF_AUDIT.md`; guard, BZero field, and WordBZero producer packets. The composition adds no new arithmetic formula.